



UDYAM'24

ANNUAL TECHNICAL FEST OF ELECTRONICS ENGINEERING SOCIETY

DIGISIM PS - 1

In a classroom far, far away, mischievous twins Karan and Kabir found themselves strategically separated by their clever teacher. Now, being the masterminds of mischief, they weren't going to let desks and distance spoil their fun. Instead, they hatched a plan to communicate secretly, turning their classroom into a cryptic battleground!

Kabir, the true genius among the two, came up with an ingenious idea – a simplified version of the DES algorithm. Here's the deal: their secret messages, a mere 8 bits long, needed an extra layer of mystery. So, they started with an initial key of 8 bits. Kabir, in his genius moment, sliced it into 4 round keys, each a cute 6 bits. Now, the real fun began. They took their innocent 8-bit message and sent it through a series of operations, mingling and jumbling it with the 4 round keys in not one, but four rounds! Each round added a dash of complexity, transforming the plain text into a final secret code that only Karan and Kabir could understand.

It's like they were crafting their own secret language right under the teacher's nose! Kabir, with a twinkle in his eye, now needs your brainpower to make this encryption even cooler, by turning this idea into a digital circuit ! Let's get into the details !!



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PART - 1 (Mandatory with 80% weightage)

Your task is to design and simulate a digital circuit that encrypts a given digital input using a simplified version of the DES algorithm that's been explained in this problem statement.

The Inputs :

- 8 bit plain text (the message to be encrypted)
- 8 bit initial key

Let us first look at the overall process of encryption , and then we will understand each part separately.

Refer to fig.1,

The 8-bit plain text is initially permuted with initial permutation table given below.

initial plain text indices	1	2	3	4	5	6	7	8
permuted plain text indices	3	4	8	7	1	2	5	6

initial permutation table

Example :

If your 8 bit input message / plain text is : 11010010

Then your permuted message will be : 01011100



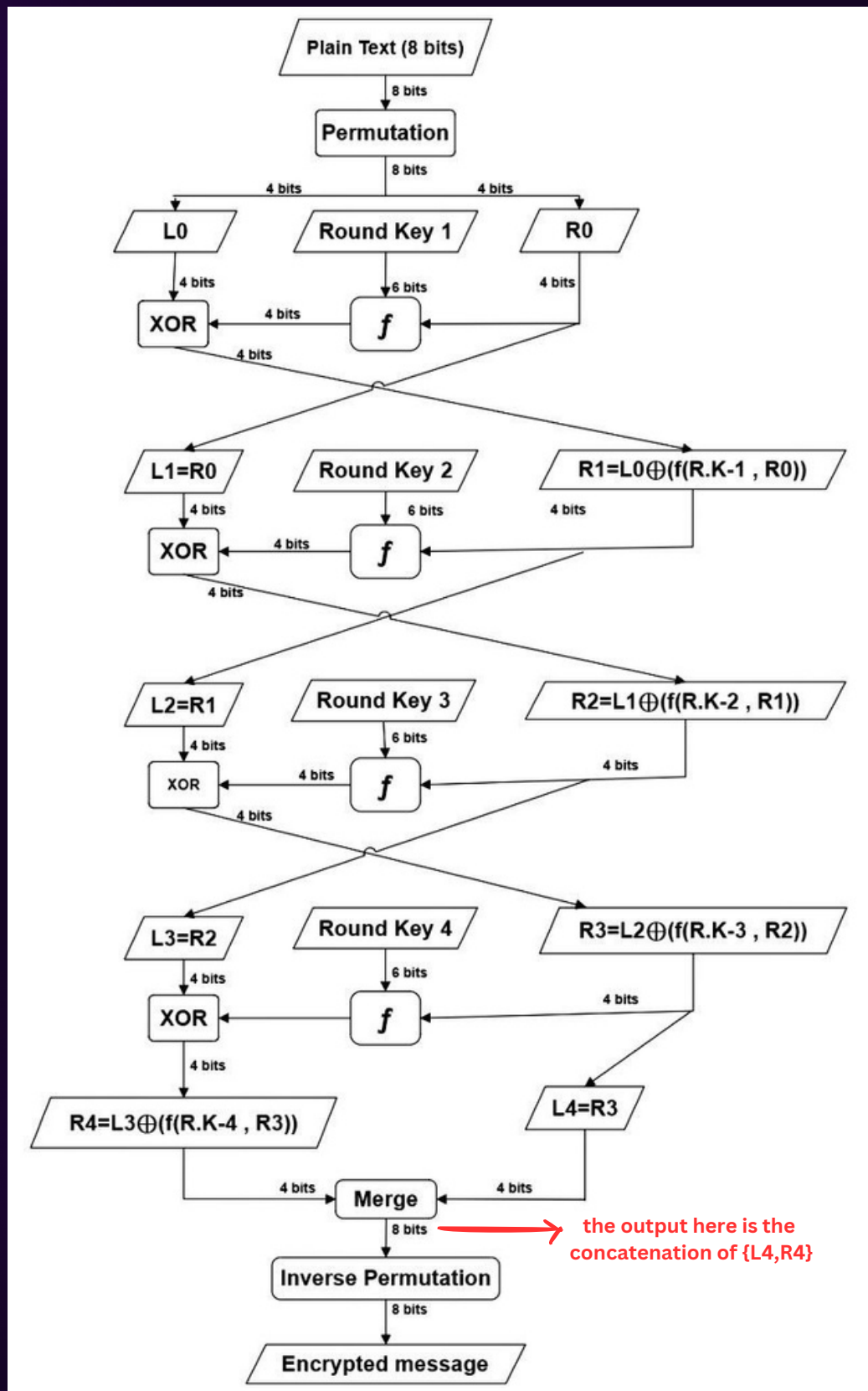


fig. 1





After the plaintext has been permuted with initial permutation table, the string is now split into two halves , the right half R_0 and the left half L_0 , each of 4 bits.

Now begin the rounds..

For round 'i', there is a round key - i ($RK-i$) that will be discussed in detail further. The right half is manipulated in every round using a function ' f ' to which the inputs are the right half R_i (4 bits) and the round key $RK-i$ (6 bits).

The output of the function f (4 bits) is XORed with the left half of the plain text , which has been unchanged.

The output of the XOR operation is now the right half for the next round. The right half of the present round is the left half for the next round. Refer to figure - 1 to understand the process in detail.

After round - 4 is completed , R_4 and L_4 are merged & inverse permuted using the below inverse permutation table :

merged o/p indices	1	2	3	4	5	6	7	8
inverse permuted indices	5	6	1	2	7	8	4	3

inverse permutation table

Example :

1 2 3 4 5 6 7 8

If your 8 bit merged o/p is : 01011100

5 6 1 2 7 8 4 3

Then your inverse permuted message will be : 11010010





Round Key Generation :

As mentioned earlier , an initial key of 8 bits will be given as input. From this key, 4 round keys are generated. Refer to fig.2

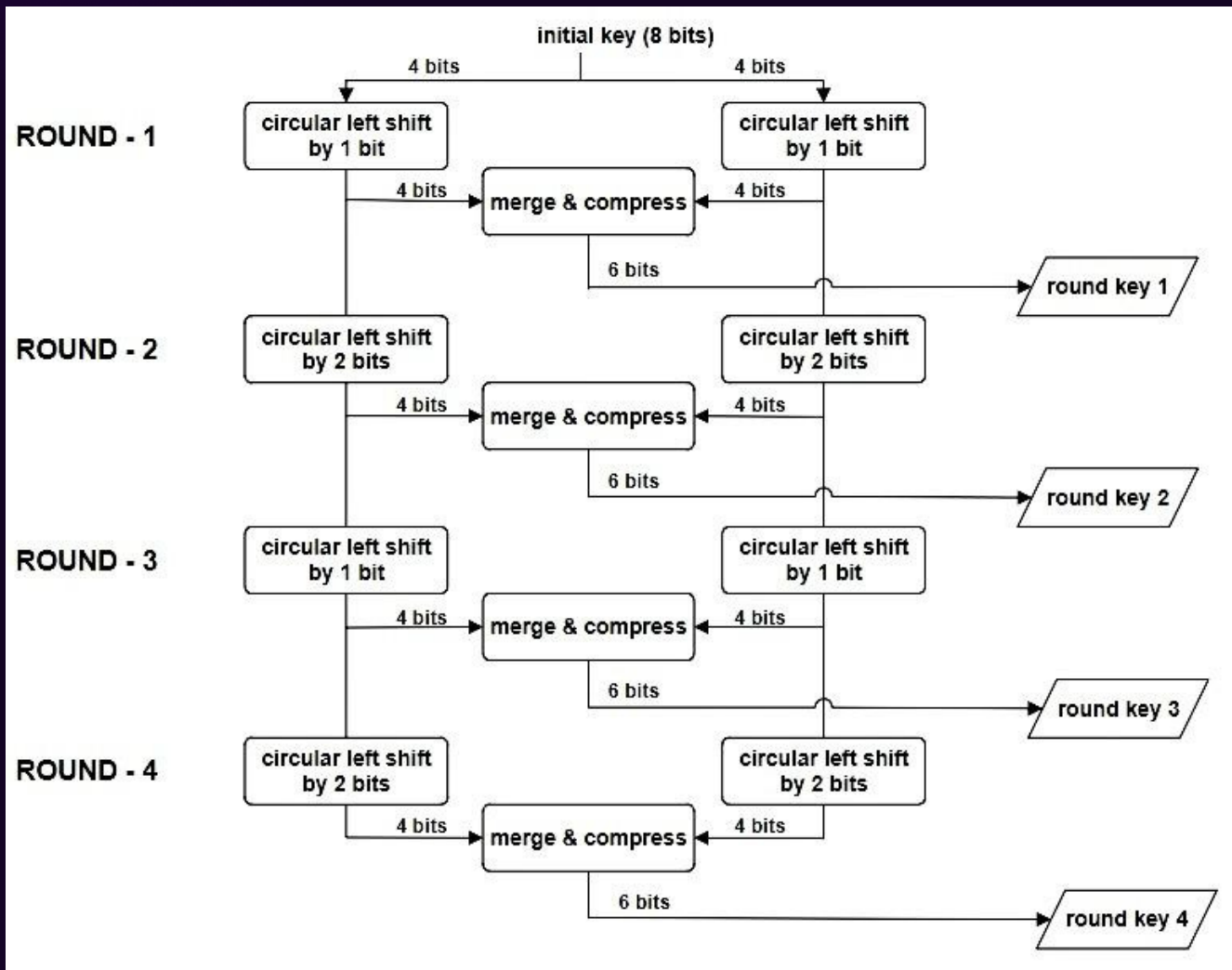


fig.2

NOTE THAT CIRCULAR LEFT SHIFT IS DONE TWICE IN EVEN ROUNDS AND ONCE IN ODD ROUNDS





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Note that the compression of 8 bits into 6 bits is done with respect to the below compression box : (It is also permuted while compression)

8 bit input indices

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

6 bit output indices

3	1	7	4	8	6
---	---	---	---	---	---

compression box

Example :

1 2 3 4 5 6 7 8

i/p : 11010010

3 1 7 4 8 6

o/p : 011100

The function f :

The round function f takes two inputs :

- (i) The right half (4 bits)
- (ii) The round key (6 bits)

The right half is first expanded (and simultaneously permuted) to 6 bits using the below expansion box

i/p indices

1	2	3	4
---	---	---	---

o/p indices

4	3	2	3	1	2
---	---	---	---	---	---

expansion box

Example :

1 2 3 4

i/p : 1101

4 3 2 3 1 2

o/p : 101011



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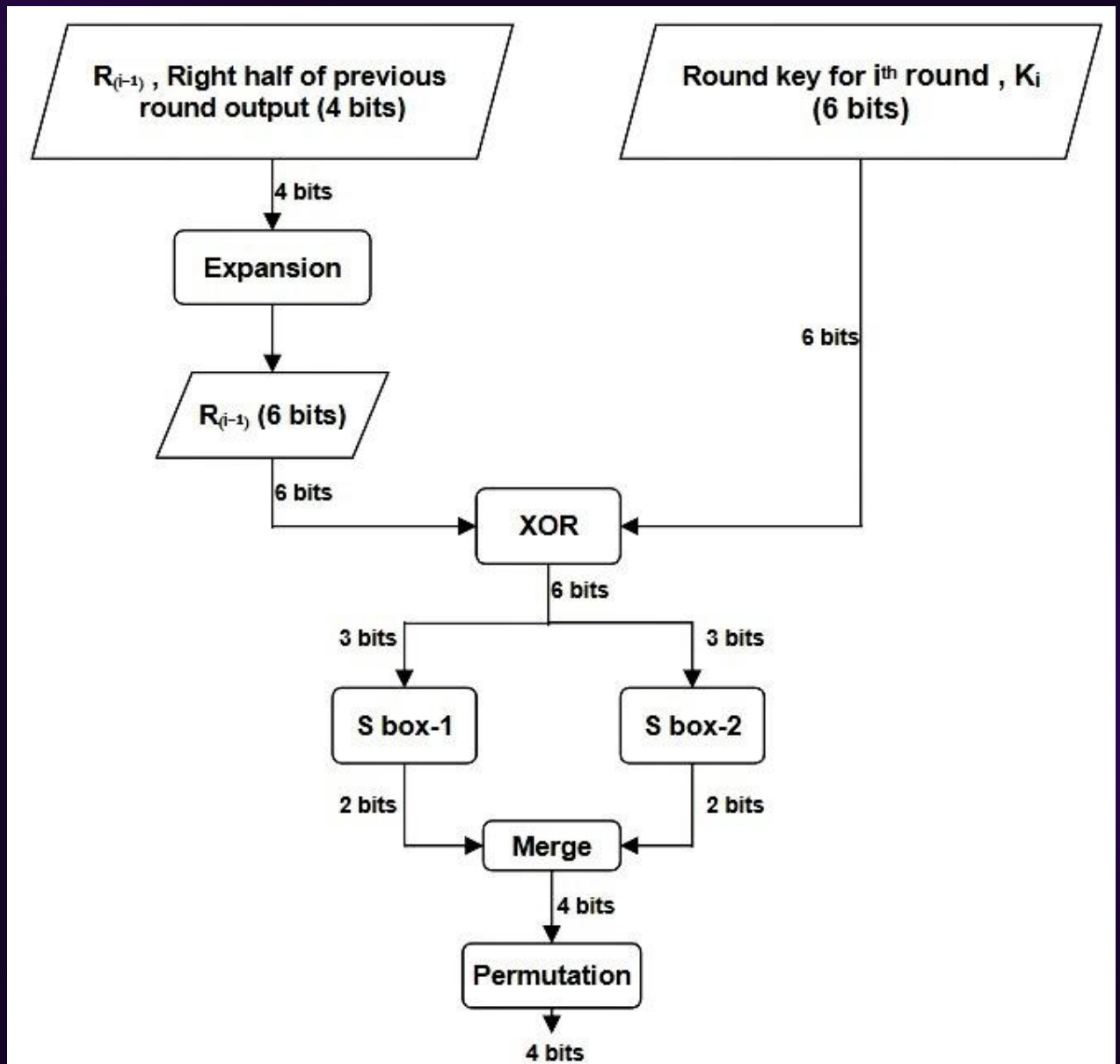


fig.3

After the XOR operation, the 6 bits are split into 3 bits each (left and right) and sent into two S boxes that each produce 2 bits.





Working of the S boxes :

The two S boxes each take 3 bit inputs and produce a 2 bit output. For a given three bit input , the middle bit represents the row number and the outer two bits represent the column number. The output of the S box is a two bit value that is present in the cell that intersects the row and column. The two S boxes are as follows.

S box - 1

	00	01	10	11
0	01	00	00	01
1	11	10	11	10

S box - 2

	00	01	10	11
0	11	10	10	01
1	00	10	10	01

Example :

If the 3 bit input to S box - 1 is 101 , it implies that :

- row number is 0
- column number is 11

The output of the S box is 01

S box - 1

	00	01	10	11
0	01	00	00	01
1	11	10	11	10





Once the S box operations are done , the two 2-bit outputs are merged and permuted according to the below permutation table :

i/p indices	1	2	3	4
o/p indices	4	1	3	2

permutation table for the round function (4 bits)

Example :

i/p : 1 2 3 4
 1001
o/p : 4 1 3 2
 1100

Now , this permuted result is the final output of the round function.

In this manner , the key generation and round function blocks are combined in a way to complete the encryption process in 4 distinct rounds.





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Example

Plaintext = 11010010
Key = 10110011

Key Generation

① Round 1 Key

Left half (LH): 1011 ; Right Half (RH): 0011
↓ Left shift (1 bit) (LS) ↓ LS

LH₁: 0111 RH₁: 0110

→ Merge →
01110110
↓ compress
RK-1: 101101

② Round Key 2

LH₂: 0111 RH₂: 0110
↓ LS (2 bit) ↓ LS (2 bit)

LH₂: 1101 RH₂: 1001
→ Merge →
11011001
↓ compress
RK-2: 010110

"Culture is the soul and the mind and the spirit." — Jawaharlal Nehru





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Round 3

$LH_2: 1101$ $RH_2: 1001$
↓ 15 (1 bit) ↓
 $LH_3: 1011$ → merge ← $RH_3: 0011$
↓
10110011
↓ compress
RK-3: 111110

Round 4

$LH_3: 1011$ $RH_3: 0011$
↓ 15 (2 bits) ↓
 $LH_4: 1110$ $RH_4: 1100$
↓ merge ↓
11101100
↓ compress
RK-4: 110001

"In a gentle way, you can shake the world." - Mahatma Gandhi



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Encryption

Plaintext: 110110010

~~Permutation~~ / Permutation

01011100

$R_0: 0101$

$R_0: 1100$

$f(R_{k-1}, R_0):$

$\rightarrow R_0 \xrightarrow{\text{Expansion}} 001011$

$\rightarrow R_0 \oplus R_{k-1} : (001011) \oplus (101101)$

100110

100

110

$\downarrow$ S-Box 1

$\downarrow$ S-Box 2

00

10

$\rightarrow$ Merge $\leftarrow$

0010

$\downarrow$ Permuted

$\therefore f(R_{k-1}, R_0) = 0010$

$R_1 = L_0 \oplus f(R_{k-1}, R_0) = 0111$

$L_1 = 1100$

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$f(RK-2, R1) :$

$R1 \xrightarrow{\text{Expression}} 111101$
(0111)

$R1 \oplus RK-2 : (111101) \oplus (010110)$
 $: 101011$

$\downarrow$
101 011
 $\downarrow$ S Box 1 $\downarrow$ S Box 2

01 $\xrightarrow{\text{Merge}}$ 10

011001
 $\downarrow$
0110

$\downarrow$ Permutatⁿ

$f(RK-2, R1) = 0011$

$R2 : L1 \oplus f(RK-2, R1)$

$: (1100) \oplus (0011)$

$: 1111$

$L2 : 0111$

$f(RK-3, R2) :$

$R2 \xrightarrow{\text{Expression}} 111111$

$R2 \oplus RK-3 = (111111) \oplus (111110)$

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000001

000 001

01 10

Merge & Permute

$f(RK-3, R2) = 0011$

$R3 = L2 \oplus f(RK-3, R2)$
 $= (0111) \oplus (0011)$
 $= 0100$

$L3 = 1111$

$f(RK-4, R3) :$
 $R3 \rightarrow 001001$
 $R3 \oplus RK-4 = (001001) \oplus (110001)$
 $= 111000$

111 000

10 11

Merge & Permute

$f(RK-4, R3) = 1110$

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$$R_4 = L_3 \oplus f(R_{K-4}, R_3)$$
$$= (1111) \oplus (1110)$$
$$= 0001$$
$$L_4 = 0100$$

L_4 R_4

 ↓ ↙ ↘

 merge

 ↓

$$01000001$$

 ↓ Inverse Permutation

Encrypted : 00010100

Data : (1100) ⊕ (1110)

0010

1111

(29, 4, 19)

100100

(10001) ⊕ (100100) = 29 ⊕ 19

000111

000

111

11

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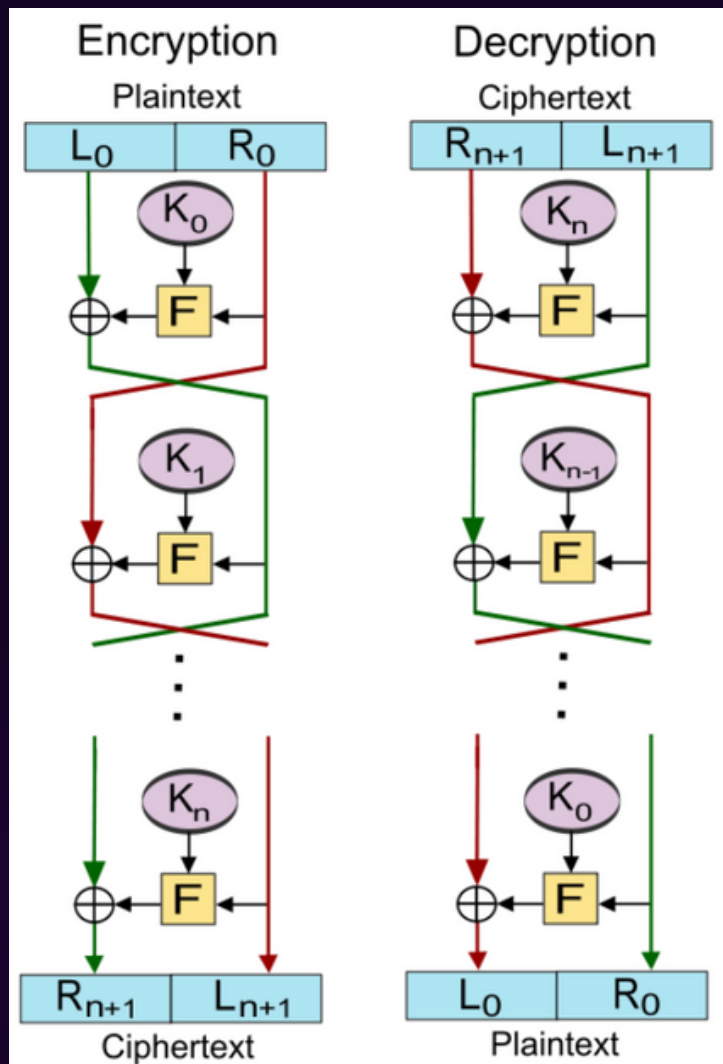
PART- 2 BONUS (20 % weightage)

Using the same circuit that you've designed for encryption , with a few changes , modify your circuit such that it works for both encryption and decryption based on the input mode . In this case you will have a total of three inputs (i. cipher text/plain text ; ii. initial key ; iii. mode)

mode = 0 ---> Encryption

mode = 1 ---> Decryption

Note : Although initial permutation and inverse permutation steps are not shown in the below diagram, they are part of both encryption and decryption.



ref

Reference

<https://medium.com/@ashutosh.awasthi.ece21/list/ps-1-3cea6b9770e6>



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COST OF COMPONENTS

NOTE:

You are allowed to use any components that come under the below mentioned categories , the cost of the components will be scaled up or down according to the number of bits with reference to the below mentioned base costs. For example , you're free to use any kind of multiplexer. If the base cost for n bit mux is 'x' , and if you're using $2n$ bit mux, then cost will be $2x$.

- Clock (Rs.40)
- Comparator (4 bit --> Rs.2)
eg : 7485, 4585, etc
- Multiplexer (Quad 2:1 mux --> Rs.2)
eg : 74157, 74153, etc
- Adder (4 bit --> Rs.2)
eg : 74283, 7482, etc
- Encoder (8 : 3 --> Rs.2)
eg : 74HC148, etc
- Decoder (4 : 16 --> Rs.2)
eg : 74LS139 , 74HC154, etc
- Flip Flops (1 bit --> Rs.1)
eg : 74273, 7474, 74LS175, 74LS109, etc
- Register (4 bit --> Rs.2)
eg : 74179, 74194, etc
- Counter (4 bit --> Rs.2)
eg : 74LS590, 74161, 74163, etc
- Logic Gates (1 or 2 i/p --> Rs 0.1 ; 3 i/p --> Rs 0.2 ; 4 i/p--> Rs 0.3..)
- Buffer (1 bit tristate buffer --> Rs 0.1)
eg : 74HC241, 74125, etc

SUBMISSION REQUIREMENTS:

You are required to submit the folder drive link consisting of :

- A PDF file containing description of your approach
- The Proteus Design file or the Project file.
- In case you do both the parts, you must submit pdf and proteus files for each part.

